

120 Days of Quant Finance

Week 3 – Linear Algebra (Finance)

Day 5: Factor Models and Portfolio Decomposition

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Introduction

Financial assets rarely move independently. Instead, their returns are driven by a small number of common sources of variation known as *factors*. Factor models provide a structured framework to explain asset returns, decompose portfolio risk, and construct scalable risk models.

This section introduces linear factor models and demonstrates how portfolio risk can be decomposed into systematic (factor) and idiosyncratic components.

Single-Asset Factor Model

For asset i , the linear factor model is written as:

$$r_i = \alpha_i + \beta_i^\top \mathbf{f} + \varepsilon_i,$$

where:

- r_i is the asset return,
- α_i is the intercept term,
- $\mathbf{f} \in \mathbb{R}^k$ is the vector of factor returns,
- $\beta_i \in \mathbb{R}^k$ represents factor loadings,
- ε_i is the idiosyncratic return component.

This model assumes asset returns are linear functions of common factors plus asset-specific noise.

Matrix Representation

Stacking all n assets, the factor model can be written compactly as:

$$\mathbf{r} = \boldsymbol{\alpha} + \mathbf{B}\mathbf{f} + \boldsymbol{\varepsilon},$$

where:

- $\mathbf{r} \in \mathbb{R}^n$ is the vector of asset returns,
- $\boldsymbol{\alpha} \in \mathbb{R}^n$ is the intercept vector,
- $\mathbf{B} \in \mathbb{R}^{n \times k}$ is the factor loading matrix,
- $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ is the idiosyncratic noise vector.

This matrix formulation enables efficient modeling of large portfolios.

Assumptions of the Factor Model

Standard factor models assume:

- Factors are uncorrelated with idiosyncratic returns,
- Idiosyncratic returns are mutually uncorrelated,
- $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$.

Under these assumptions, factor models yield a tractable covariance structure.

Covariance Decomposition

Let:

- $\boldsymbol{\Sigma}_f = \text{Cov}(\mathbf{f})$ denote factor covariance,
- $\boldsymbol{\Sigma}_\varepsilon = \text{Cov}(\boldsymbol{\varepsilon})$ be a diagonal matrix of idiosyncratic variances.

The asset return covariance matrix decomposes as:

$$\boldsymbol{\Sigma} = \mathbf{B}\boldsymbol{\Sigma}_f\mathbf{B}^\top + \boldsymbol{\Sigma}_\varepsilon.$$

This decomposition separates systematic and idiosyncratic sources of risk.

Portfolio Risk Decomposition

Let $\mathbf{w} \in \mathbb{R}^n$ denote the portfolio weight vector. Portfolio variance is given by:

$$\text{Var}(R_p) = \mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{w}.$$

Substituting the factor covariance structure:

$$\text{Var}(R_p) = \mathbf{w}^\top \mathbf{B} \boldsymbol{\Sigma}_f \mathbf{B}^\top \mathbf{w} + \mathbf{w}^\top \boldsymbol{\Sigma}_\varepsilon \mathbf{w}.$$

The two terms represent:

- **Factor risk** (systematic risk),
- **Idiosyncratic risk** (diversifiable risk).

Portfolio Factor Exposures

The portfolio's exposure to factors is given by:

$$\boldsymbol{\beta}_p = \mathbf{B}^\top \mathbf{w}.$$

This vector summarizes how the portfolio loads on each underlying factor.

Statistical and Economic Factor Models

Factor models can be broadly classified into:

- **Statistical factor models**, where factors are extracted via methods such as PCA and are orthogonal by construction.
- **Economic factor models**, where factors are pre-specified based on economic intuition or market structure.

Statistical factors are data-driven but less interpretable, while economic factors offer interpretability at the cost of potential model misspecification.

PCA as a Factor Model

Let $\boldsymbol{\Sigma}$ denote the covariance matrix of asset returns. PCA provides the eigen-decomposition:

$$\boldsymbol{\Sigma} = \mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^\top.$$

In this representation:

- Columns of $\mathbf{Q}$ act as factor loading vectors,
- Diagonal elements of $\mathbf{\Lambda}$ represent factor variances.

This corresponds to a purely statistical factor model.

Interpretation and Limitations

Factor models explain how risk is distributed across common drivers, but they do not guarantee return predictability. Key limitations include:

- Time-variation of factor loadings,
- Instability of statistical factors,
- Sensitivity to model assumptions.

Factor models should therefore be used primarily for risk analysis rather than alpha generation.

Conclusion

Factor models provide a powerful framework for decomposing portfolio risk into systematic and idiosyncratic components. By representing portfolios in factor space, quantitative researchers gain insight into the true drivers of risk and can design more robust portfolios and risk controls.

Understanding factor models is essential for portfolio construction, risk management, and advanced quantitative research.